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B.Tech. III Sem.

BEM – C302 (Engineering mathematics – III)

All Branches

Date: 26.09.2025

Time: 2:00-3:00

Sec. A: - Attempted any 2 questions. M.M. – 6 X 2

1 9. If $F(z) = \frac{2z^2+3z+12}{(z-1)^4}$ find the value of $f(1)$ and $f(2)$.

2 10. Find the inverse Z-transform of the function $\frac{z}{(z-2)(z-3)^2}$

3 11. Find the positive root of the equation $x^3 - 2x - 5 = 0$ correct to four decimal places, using Newton- Raphson method.

4 12. Find a real root of the equation $\cos x = 3x - 1$ correct to three decimal places using Iteration method.

Sec. B: - Attempt any 1 question. M.M. – 8 X 1

1 5. Solve $y_{n+2} + 4y_{n+1} + 3y_n = 3^n$; $y_0 = 0, y_1 = 1$

2 6. Find the root of the equation $xe^x = \cos x$ correct to four decimal places by using Secant method.

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BEM–C302

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All Branches (SE, ECE, EE, ME)

Date:21.11.2025

Time:02:00-03:00

...:-Attemptedany2 questions. M.M.–6X2	Sec.B:-Attemptany1 question. M.M.–8X1
...1: Define analytic function. Also, show that the function $f(z) = z^2$ is an analytic function. Q.2:Find the value of c_1 and c_2 such that the function $f(z) = x^2 + c_1y^2 - 2xy + i(c_2x^2 - y^2 + 2xy)$ is analytic. Q.3:Evaluate $\int_0^{1+i}(x^2 - iy)dz$ along the path (a) $y = x$ (b) $y = x^2$ Q.4:Determine thepoles and residue of the function $f(z) = \frac{z^2}{(z-1)^2(z+2)}$	Q.1: Define harmonic function. If $u(x, y) = 3x - 2xy$, then show that $u(x, y)$is harmonic function. Also find $v(x, y)$such that $f(z) = u + iv$ is an analytic function and then express $f(z)$ in terms of z. Q.2:Evaluate $\int_C \frac{12z-7}{(z-1)^2(2z+3)} dz$ where C is the circle $\|z\| = 3$.